

PacketLens

PCAP Network Analysis Report Tool — User Guide

Version 3.2.0 · Report schema 1.0 · Risk specification 1.3
Web application (self-hosted Next.js server)

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1. What is PacketLens?

PacketLens is a network security analysis tool that turns a raw packet capture (PCAP / PCAPNG) into a professional, exportable security report. Upload a capture file — or open the built-in demo dataset — and PacketLens parses the traffic on the server, then produces a 22-section report covering traffic summary, packets, flows, sessions, DNS, HTTP, TLS, VoIP calls, extracted files, credentials, certificates, endpoints, timeline, top talkers, alerts, indicators of compromise (IOCs), MITRE ATT&CK mappings, a numerical risk score, and prioritized recommendations.

PacketLens is a single web application: one Next.js server that serves the interface, accepts uploads, runs the analysis, and serves the results back to your browser. There is no separate desktop app and no separate analysis engine to deploy — the capture file is processed entirely by the server you run.

2. Quick Start

2.1 Running the app

1. Install Node.js 20 or newer.
2. Run `npm install` inside the `frontend/` directory.
3. Development: run `start-dev.bat` (Or `npx next dev --port 3456`) and open <http://localhost:3456>.
4. Production: `npm run build` , then `npm start` (see section 10).

2.2 First steps

1. Click **Live Demo** on the landing page to explore every feature with a realistic pre-loaded dataset — no capture file needed.
2. Or drop a PCAP / PCAPNG file onto the upload zone.
3. The analysis opens automatically when parsing finishes. Bookmark the job URL to return to the same analysis later.

3. Analyzing a Capture

- **Upload a file** — PCAP and PCAPNG formats, up to 500 MB. Uploads are rate-limited (5 per minute per address) and rejected from other origins. While analysis runs, a progress screen shows the decode rate and current stage.
- **Demo dataset** — a realistic capture with benign + malicious traffic, pre-loaded with analysis, so every feature can be explored without a capture file. Demo analyses are labelled as such in the report.
- **Encapsulation** — Ethernet is fully supported. Unsupported link types (e.g. raw IP) still produce packet lengths and timestamps; the report clearly states the decode rate and withholds a verdict when too little traffic was decodable (it reads UNKNOWN / INSUFFICIENT DATA rather than a false "clean" score). An explicit link-layer override can be set before uploading for captures whose headers were not auto-detected.

- **History** — completed analyses are stored on the server in `~/.packetlens/jobs.json`; the 20 most recent are kept, with the oldest evicted automatically. A bookmark to the job URL reloads the full analysis.

The capture file is processed on the server hosting this deployment, never in your browser. It is held in memory during the request and is not written to disk; only the finished analysis result is persisted (see the Privacy Policy).

4. The Analysis Report

Every analysis produces one canonical report with 22 numbered sections:

- | | |
|----------------------|-------------------------------------|
| 1. Executive Summary | 12. Certificates |
| 2. Traffic Summary | 13. Endpoints |
| 3. Packets | 14. Threats & Alerts |
| 4. Flows | 15. Timeline |
| 5. Sessions | 16. Top Talkers |
| 6. DNS Analysis | 17. Risk Score |
| 7. HTTP Analysis | 18. Indicators of Compromise (IOCs) |
| 8. TLS Analysis | 19. MITRE ATT&CK Mapping |
| 9. Extracted Files | 20. Recommendations |
| 10. VoIP / SIP Calls | 21. Appendix |
| 11. Credentials | 22. Analyst Conclusion |

Sections cover packet counts, flows, sessions, local devices, DNS lookups (deduplicated through the LAN resolver), plaintext HTTP requests, TLS handshakes and certificate chains, files reconstructed from HTTP traffic, per-destination country distribution via the installed GeoIP database, credentials spotted in cleartext auth traffic (masked by default, revealable per entry), VoIP calls with RTP statistics, TCP health (handshake RTT, retransmissions, out-of-order, zero-window, RST, loss), and a per-interval activity timeline.

A **Beginner / Expert** toggle adjusts how much detail the report shows.

5. Exporting Results

From the report page, three export formats are available:

Format	What you get	Best for
PDF	The full report via your browser's print dialog: cover page with file metadata and a demo/live marker, all 22 sections, print-optimized tables that avoid page-break splitting.	Sharing with stakeholders, evidence files.
HTML	A standalone, self-contained HTML file with the full summary, tables, and a deep link back to the live report in PacketLens. Works offline, styled for reading.	Email attachments, archiving, viewing in any browser.
CSV	One row per network flow: IPs and ports as separate columns (IPv6-safe), sent/recv bytes, timing, duration, geo country/ASN per endpoint, service name, RTT, retransmissions, loss. UTF-8 BOM included so Excel/Sheets decode vendor names correctly.	SIEM ingestion, pandas/Excel pivot analysis.

6. Interactive Tools

6.1 Investigation Graph

A cytoscape-based graph of the capture's communication: local devices and remote endpoints as nodes, conversations as edges. Click a node to see its traffic, drag to rearrange, zoom, and filter. Useful for spotting star-pattern C2 or broadcast storms at a glance. Layout and display preferences persist in your browser.

6.2 World Map

A geographic view plotting every remote endpoint by country with arcs representing traffic flows, animated to show conversation volume, and alert markers for flagged hosts. Pins are sized by traffic volume; nearby endpoints cluster into city-level bubbles. Works without the GeoIP database — it simply degrades to "no location data" notes. Without a home location the map anchors arcs at a synthetic point; set Home in Settings for exact lines.

7. Detection & Risk Model

- **Signature rules** — port scanning, SYN floods, DNS tunneling, credential leaks, malware downloads, C2 beaconing, data exfiltration, suspicious TLS, and cleartext HTTP credentials. Each alert carries severity, confidence, source/destination, and evidence.
- **Behavioral metrics** — beaconing periodicity, DNS tunneling volume, data exfiltration egress, port-scan breadth, JA3 TLS fingerprint anomalies, and traffic bursts. These feed risk contributions and, when proven, also fire as real signature alerts.
- **Risk score** — a single 0–100 score computed from a transparent formula ($100 \times (1 - e^{(-raw/80)})$): each alert contributes (severity weight + rule weight) $\times$ a confidence multiplier, with burst detection able to boost eligible rules. The report shows the full breakdown per rule, and always states the spec version used.
- **IOCs & MITRE** — every finding is classified (IP, domain, URL, certificate...) and mapped to MITRE ATT&CK techniques (T1046, T1048, T1071, T1552, etc.) with the source labeled as confirmed alert or behavioral indicator.

- **Honest verdicts** — if a capture's encapsulation prevented header decoding, the verdict reads UNKNOWN / INSUFFICIENT DATA instead of a false "clean" score.

8. GeoIP & Vendor Attribution

- **GeoIP database** — the first time a capture needs country attribution, the server downloads the free DB-IP City Lite database (≈125 MB, CC BY 4.0) once and stores it in `~/.packetlens/geo/`. The install is single-flight: concurrent requests share one download.
- **Custom database** — you can instead upload your own MaxMind-format MMDB file via Settings → GeoIP Database; the uploaded file replaces the auto-installed one. Remove it anytime with the Remove button (auto-install resumes on the next analysis).
- **Online lookups (optional)** — Online GeoIP Lookups is disabled by default. When enabled, only IP addresses missing from the local database are resolved via the ipwho.is service; nothing else is sent.
- **Vendor lookup** — MAC-address vendors come from the bundled OUI registry (`oui.json`) and never require a network call. Raw-IP captures without Ethernet headers have no MACs, so vendor lookup reports "no MAC addresses" instead of a fake 0.

9. Settings

Setting	What it does
Beginner Mode	Simplifies labels and masks IP addresses in the graph view.
Link-layer override	Forces a DLT (e.g. Ethernet, Raw IP, Linux SLL) when the capture's encapsulation is not auto-detected.
Manual Home Location	Lat/lon used to anchor the world map's local arcs. Stored in your browser only.
Online GeoIP Lookups	Opt-in ipwho.is resolution for IPs missing from the local database (off by default).
GeoIP Database	Shows install status, installs/removes the DB-IP database, and accepts an uploaded MMDB file.
Danger Zone → Clear Analysis	Discards the currently loaded analysis from the session. Server-side job retention is bounded by the 20-job cap described in section 3.

10. Self-Hosting & Deployment

PacketLens is a standard Next.js application (`output: "standalone"`):

1. `npm run build` inside `frontend/`, then `npm start` — the server listens on the configured port.
2. Persistent data lives in `~/.packetlens/` (analysis history `jobs.json`, GeoIP database `geo/`). Mount a persistent volume there (or set `HOME`) so jobs and the database survive restarts.

3. Memory: an upload is fully buffered and analyzed in-process (500 MB cap), so concurrent uploads need roughly 1.5–2.5 GB each. Size the instance accordingly (4 GB RAM is comfortable).
4. Behind a reverse proxy, forward the client address (e.g. `X-Real-IP`) so the upload rate limit keys on real clients. TLS terminates at the proxy; the app already emits HSTS and CSP headers.

PacketLens is designed as a single instance: analysis history and rate limiting live in one process. Horizontal scaling would require externalizing the job store and rate limiter.

11. Security Notes

- Uploads are restricted to .pcap/.pcapng, capped at 500 MB (checked against Content-Length before the body is buffered), rate-limited per client address, and rejected from cross-origin callers.
- File names are sanitized before storage; analysis jobs are kept under a 0700 directory with 0600 file permissions.
- The app emits a restrictive Content-Security-Policy, HSTS, X-Frame-Options DENY, and nosniff headers. No third-party analytics or telemetry are embedded.
- Captures are a privacy-sensitive artifact: they are processed only on the server you run and are never written to disk. Report data (credentials, certificates, extracted files) is displayed in the app; exporting it deliberately is your choice as the analyst.
- GeoIP and OUI vendor data are open-data (DB-IP Lite CC BY 4.0 / OUI registry) and are only used for attribution labels.

12. Requirements

Item	Requirement
Server	Node.js 20+, ~2 GB free memory (4 GB recommended for concurrent uploads), disk for <code>~/.packetlens</code> (GeoIP database ≈125 MB)
Browsers	Chrome, Edge, Firefox, Safari (recent versions)
Network (server)	Outbound HTTPS for the optional one-time GeoIP database download and the opt-in ipwho.is lookups